
LINUX COMMANDS COMPLETE REFERENCE

547+ commands across 18 categories

File Management • Networking • Security • Process Control • Development • and more

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■ FILE & DIRECTORY MANAGEMENT

20 commands

ls	List directory contents. Shows files and subdirectories in the current or specified directory.
cd	Change directory. Moves the shell's working directory to the specified path.
pwd	Print working directory. Displays the full absolute path of the current directory.
mkdir	Make directory. Creates one or more new directories at the specified path(s).
rmdir	Remove directory. Deletes empty directories; fails if the directory contains files.
rm	Remove files or directories. Permanently deletes files; use -r to remove directories recursively.
cp	Copy files or directories. Duplicates source to destination; use -r for directories.
mv	Move or rename files and directories. Relocates or renames a file/directory.
touch	Create an empty file or update the access/modification timestamps of an existing file.
find	Search for files in a directory hierarchy based on name, type, size, permissions, and more.
locate	Find files by name using a pre-built database index. Much faster than find for simple lookups.
tree	Display directory contents in a tree-like format, showing the hierarchy visually.
ln	Create hard or symbolic (soft) links between files. -s creates a symbolic link.
realpath	Print the resolved absolute path of a file, following all symlinks.
stat	Display detailed metadata about a file or filesystem: size, permissions, timestamps, inode.
file	Determine the type of a file by examining its content rather than its extension.
basename	Strip directory and optionally suffix from a file path, returning just the filename.
dirname	Return the directory component of a file path, stripping the filename.
rename	Rename multiple files at once using Perl-style regular expressions.
shred	Securely delete a file by overwriting its contents multiple times before removal.

■ FILE VIEWING & EDITING

38 commands

cat	Concatenate and display file contents. Also used to combine multiple files into one.
less	View file contents one screenful at a time with backward/forward navigation.
more	View file contents page by page; unlike less, it only allows forward navigation.
head	Display the first N lines of a file (default 10). Useful for previewing large files.
tail	Display the last N lines of a file. -f follows a file live as it grows (e.g., logs).
nano	Simple, beginner-friendly terminal text editor with on-screen keybinding hints.
vim	Powerful modal text editor with extensive customization, macros, and plugin support.
vi	Original visual text editor; vi-compatible subset available on virtually all Unix systems.
emacs	Extensible, self-documenting text editor with built-in Lisp interpreter and vast ecosystem.
gedit	GNOME graphical text editor for desktop environments.
diff	Compare two files line by line and show the differences between them.
patch	Apply a diff (patch) file to update a file or source tree to a newer version.
wc	Word count: count lines, words, and bytes/characters in a file or standard input.

sort	Sort lines of text in a file or from stdin alphabetically, numerically, or by field.
uniq	Filter adjacent duplicate lines from sorted input; also counts or only shows duplicates.
cut	Extract specific columns or fields from each line of a file, delimited by a character.
paste	Merge corresponding lines from multiple files side by side, separated by a delimiter.
join	Join lines from two sorted files on a common field, similar to a database join.
tr	Translate, squeeze, or delete characters from standard input. E.g., convert case.
sed	Stream editor: perform text transformations (substitutions, deletions, insertions) on files.
awk	Powerful text-processing language that operates on records and fields within files.
grep	Search for lines matching a pattern (regex) in files or standard input.
egrep	Extended grep: same as grep -E, supports extended regular expressions.
fgrep	Fixed-string grep: same as grep -F, treats the pattern as a literal string (faster).
strings	Extract printable strings from binary files, useful for basic binary analysis.
od	Octal dump: display file contents in octal, hex, decimal, or ASCII representations.
xxd	Create a hex dump of a file or convert a hex dump back to binary.
nl	Number the lines of a file and write to standard output.
fold	Wrap lines in a file to fit a specified width.
expand	Convert tab characters in a file to the equivalent number of spaces.
unexpand	Convert leading spaces back to tab characters in a file.
col	Filter reverse line feeds and other special characters from terminal output.
colrm	Remove specific columns from each line of a text file.
column	Format input into aligned columns; useful for making tabular data readable.
fmt	Simple text formatter that fills and joins lines to a specified width.
pr	Paginate or columnate files for printing, adding headers and page numbers.
tac	Reverse cat: display file contents in reverse line order (last line first).
rev	Reverse the characters on each line of a file or standard input.

■ FILE PERMISSIONS & OWNERSHIP

8 commands

chmod	Change file mode (permissions): read, write, execute for owner, group, and others.
chown	Change the owner and/or group of a file or directory.
chgrp	Change only the group ownership of a file or directory.
umask	Set the default permission mask applied when new files or directories are created.
getfacl	Get file access control lists (ACLs), showing extended permission settings.
setfacl	Set file access control lists, allowing fine-grained permissions for specific users/groups.
lsattr	List file attributes on an ext2/ext3/ext4 filesystem (immutable, append-only, etc.).
chattr	Change special filesystem attributes on a file (e.g., make a file immutable).

■ FILE COMPRESSION & ARCHIVING

15 commands

tar	Tape archive: bundle files into a single archive (.tar). Add -z/-j for compression.
gzip	Compress files using the gzip algorithm (.gz). Use gunzip to decompress.
gunzip	Decompress files compressed with gzip.
bzip2	Compress files using the bzip2 algorithm (.bz2), generally better compression than gzip.
bunzip2	Decompress files compressed with bzip2.
xz	Compress files using the LZMA/xz algorithm (.xz), offering very high compression ratios.
unxz	Decompress files compressed with xz.
zip	Package and compress files into a .zip archive compatible with Windows and other OSes.
unzip	Extract files from a .zip archive.
7z	Work with 7-Zip archives (.7z) and many other formats with high compression.
zcat	Display the contents of a gzip-compressed file without decompressing it to disk.
zless	View a gzip-compressed file with less without fully decompressing it.
bzcat	Display the contents of a bzip2-compressed file without decompressing to disk.
rar	Create RAR archives (proprietary format, requires non-free package).
unrar	Extract files from RAR archives.

■ PROCESS MANAGEMENT

28 commands

ps	Report a snapshot of current processes. ps aux shows all processes with details.
top	Interactive real-time process viewer showing CPU, memory usage, and system stats.
htop	Enhanced, colorful, interactive process viewer with mouse support (install separately).
kill	Send a signal to a process by PID. Default signal is SIGTERM (15) to request termination.
killall	Kill all processes matching a name. More convenient than finding individual PIDs.
pkill	Signal processes based on name or other attributes (regex support).
pgrep	Find the PIDs of running processes by name or attributes.
nice	Start a process with a modified scheduling priority (niceness from -20 to 19).
renice	Change the scheduling priority of an already-running process.
nohup	Run a command immune to hangup signals; keeps it running after you log out.
bg	Resume a suspended job in the background.
fg	Bring a background or suspended job to the foreground.
jobs	List active jobs (background/stopped processes) in the current shell session.
wait	Wait for a background process or job to complete before continuing.
sleep	Pause execution for a specified number of seconds (or minutes/hours/days).
timeout	Run a command with a time limit; kills it if it runs too long.
watch	Repeatedly execute a command at a fixed interval and display the output.
at	Schedule a command to run once at a specific time in the future.
cron	Daemon that runs scheduled tasks (cron jobs) defined in a crontab file.
crontab	Edit, list, or remove the cron job schedule for the current user.

systemctl	Control the systemd system and service manager: start, stop, enable, disable services.
service	Start, stop, restart, or check the status of a SysV init or systemd service.
journalctl	Query and display logs from the systemd journal.
strace	Trace system calls and signals made by a process; invaluable for debugging.
ltrace	Trace library calls made by a process (calls to shared libraries like libc).
lsof	List open files and the processes that have opened them; includes network sockets.
fuser	Identify processes using a file, directory, or socket.
pmap	Show memory map of a process, detailing memory regions and their usage.

■ SYSTEM INFORMATION

24 commands

uname	Print system information: kernel name, version, machine hardware, and OS.
hostname	Show or set the system's hostname (network name).
hostnamectl	Query or change the system hostname and related settings (systemd systems).
uptime	Show how long the system has been running, current time, users, and load averages.
date	Display or set the system date and time.
timedatectl	Query and change the system clock, timezone, and NTP synchronization settings.
cal	Display a calendar for the current month or a specified month/year.
whoami	Print the username of the currently logged-in user.
who	Show who is logged in to the system and from where.
w	Show who is logged in and what they are doing (processes running).
id	Print the UID, GID, and group memberships of the current or specified user.
last	Show the history of user logins and system reboots from /var/log/wtmp.
lastlog	Show the most recent login for each user on the system.
lscpu	Display detailed CPU architecture information: cores, threads, caches, flags.
lsmem	Show the ranges of available memory and their state.
lshw	List detailed hardware configuration of the machine.
lspci	List all PCI devices (graphics cards, network adapters, etc.).
lsusb	List all USB devices currently connected to the system.
dmidecode	Decode DMI/SMBIOS data to show hardware information from BIOS.
dmesg	Print or control the kernel ring buffer (boot messages, hardware errors).
hwinfo	Probe for and display detailed hardware information (may need installation).
inxi	Comprehensive system/hardware information tool (may need installation).
arch	Print the machine hardware architecture (e.g., x86_64, aarch64).
nproc	Print the number of available processing units (logical CPUs).

■ DISK & FILESYSTEM

31 commands

df	Report disk space usage for filesystems. -h gives human-readable sizes.
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du	Estimate file space usage for files and directories. -sh gives a summary.
mount	Mount a filesystem (attach a storage device to the directory tree).
umount	Unmount a mounted filesystem (detach it from the directory tree).
fdisk	Partition table manipulator for MBR disks: create, delete, modify partitions.
gdisk	Interactive partition table manipulator for GPT disks.
parted	Powerful disk partitioning tool supporting MBR and GPT.
mkfs	Build (format) a Linux filesystem on a device (e.g., mkfs.ext4, mkfs.xfs).
fsck	Check and repair a Linux filesystem; should be run on unmounted partitions.
e2fsck	Check and repair an ext2/ext3/ext4 filesystem.
tune2fs	Adjust tunable filesystem parameters on ext2/ext3/ext4 filesystems.
blkid	Locate or print block device attributes: UUID, filesystem type, label.
lsblk	List block devices (disks, partitions, loop devices) in a tree format.
dd	Convert and copy a file, often used for creating disk images or writing ISOs.
sync	Flush filesystem buffers: force pending writes to disk.
swapoff	Disable a swap space (file or partition).
swapon	Enable a swap space to extend virtual memory.
mkswap	Set up a Linux swap area on a device or file.
btrfs	Manage Btrfs filesystems: subvolumes, snapshots, balancing, scrubbing.
zfs	Manage ZFS pools and datasets (requires ZFS kernel module).
lvdisplay	Display information about LVM logical volumes.
vgdisplay	Display information about LVM volume groups.
pvddisplay	Display information about LVM physical volumes.
lvcreate	Create a logical volume in an existing LVM volume group.
vgcreate	Create a new LVM volume group from one or more physical volumes.
pvcreate	Initialize a disk or partition as an LVM physical volume.
cryptsetup	Manage disk encryption with dm-crypt/LUKS (create, open, close encrypted volumes).
badblocks	Search a device for bad (unreadable) blocks.
hdparm	Get or set hard disk parameters, including performance settings and security features.
smartctl	Control and monitor SMART-capable hard disks and SSDs; check disk health.
ncdu	NCurses-based interactive disk usage analyzer.

■ NETWORKING

49 commands

ip	Show and manipulate routing, network devices, interfaces, and tunnels (modern replacement for ifconfig).
ifconfig	Configure network interfaces: assign IP addresses, bring up/down interfaces (legacy).
ping	Send ICMP echo requests to a host to test reachability and measure round-trip time.
ping6	Send ICMPv6 echo requests to test IPv6 connectivity.

traceroute	Trace the network path (hops) packets take to reach a destination host.
tracepath	Trace path to a host, discovering MTU along the way (no root required).
mtr	Combines ping and traceroute into a real-time network diagnostic tool.
nslookup	Query DNS servers interactively to resolve hostnames and IP addresses.
dig	DNS lookup utility: perform detailed DNS queries and display full server responses.
host	Simple DNS lookup command that resolves hostnames to IP addresses and vice versa.
whois	Query WHOIS databases to find domain registration and IP ownership information.
curl	Transfer data from/to URLs using many protocols (HTTP, HTTPS, FTP, etc.).
wget	Non-interactive network downloader; retrieve files from the web recursively.
netstat	Display network connections, routing tables, interface statistics (legacy, use ss).
ss	Socket statistics: show listening and established connections (modern netstat).
nmap	Network mapper: scan hosts for open ports, running services, and OS detection.
arp	View and manipulate the ARP cache (IP-to-MAC address mappings).
route	Show and manipulate the IP routing table (legacy; use ip route).
iptables	Configure IPv4 packet filtering, NAT, and firewall rules in the Linux kernel.
ip6tables	Configure IPv6 packet filtering and firewall rules.
ufw	Uncomplicated Firewall: user-friendly frontend for iptables rule management.
firewall-cmd	Interface to the firewalld dynamic firewall manager.
tcpdump	Capture and analyze network packets on an interface; a CLI packet analyzer.
wireshark	Graphical network protocol analyzer (CLI version: tshark).
tshark	Terminal-based Wireshark: capture and analyze network traffic from the command line.
nc	Netcat: read/write data across TCP/UDP connections; the 'Swiss Army knife' of networking.
socat	Bidirectional data relay between two data streams (more powerful than nc).
ssh	Secure Shell: open an encrypted remote login session or execute remote commands.
scp	Securely copy files between hosts over an SSH connection.
sftp	Interactive secure file transfer over SSH.
rsync	Fast, versatile file synchronization and transfer tool (local or remote via SSH).
ftp	File Transfer Protocol client for transferring files to/from FTP servers (unencrypted).
telnet	Connect to a remote host via Telnet (unencrypted; avoid for production, use for testing).
nmcli	NetworkManager command-line interface: manage connections, devices, and Wi-Fi.
nmtui	Text-based UI for NetworkManager (interactive TUI).
iwconfig	Configure wireless network interface parameters (legacy, use iw).
iw	Show or manipulate wireless devices and their configuration.
iwlist	Get detailed wireless information from a network interface (scan, frequency, etc.).
hostapd	Host access point daemon: turn a wireless interface into a Wi-Fi access point.
dhclient	DHCP client: request IP address assignment from a DHCP server.
nftables	Next-generation packet filtering framework and replacement for iptables.

vnstat	Network traffic monitor that keeps a log of network statistics.
iftop	Display real-time bandwidth usage on a network interface by connection.
nethogs	Show real-time network usage per process (which program is using bandwidth).
ethtool	Query and control network driver and hardware settings.
iperf3	Measure maximum achievable network bandwidth between two hosts.
ab	Apache HTTP server benchmarking tool: measure web server performance.
openssl	Toolkit for TLS/SSL protocols and cryptography: generate keys, certificates, encrypt files.
gpg	GNU Privacy Guard: encrypt, decrypt, sign, and verify files using public-key cryptography.

■ USER & GROUP MANAGEMENT

22 commands

useradd	Create a new user account with default or specified settings.
usermod	Modify an existing user account (change name, groups, home directory, shell, etc.).
userdel	Delete a user account and optionally their home directory and mail spool.
passwd	Change a user's password or set password aging policies.
groupadd	Create a new group on the system.
groupmod	Modify an existing group (rename, change GID, etc.).
groupdel	Delete a group from the system.
gpasswd	Administer /etc/group and /etc/gshadow: add/remove users from groups, set passwords.
su	Switch user: open a shell as another user, defaulting to root.
sudo	Execute a command as the superuser or another user, as permitted by /etc/sudoers.
visudo	Safely edit the /etc/sudoers file with syntax checking.
chpasswd	Update user passwords in batch from standard input.
newgrp	Change the current group ID during a login session.
groups	Print the names of the groups a user belongs to.
finger	Display information about system users (name, login, shell, idle time).
chsh	Change the login shell for a user.
chfn	Change user information (full name, phone numbers) stored in /etc/passwd.
vipw	Safely edit the /etc/passwd file with locking.
vigr	Safely edit the /etc/group file with locking.
lid	List the groups a user is a member of, or users who belong to a group.
faillock	Display or reset failed authentication attempt records.
pam_tally2	Display or reset the login failure count for a user (PAM module tool).

■ PACKAGE MANAGEMENT

22 commands

apt	Advanced Package Tool: install, update, and remove packages on Debian/Ubuntu.
apt-get	Lower-level Debian package management tool; predates apt but still widely used.
apt-cache	Query the APT package cache for information about packages.

dpkg	Low-level Debian package manager: install, remove, and query .deb packages.
snap	Install and manage Snap packages (containerized, cross-distro application packages).
flatpak	Install and manage Flatpak applications in sandboxed environments.
dnf	Dandified YUM: package manager for Fedora/RHEL/CentOS (modern replacement for yum).
yum	Yellowdog Updater Modified: package manager for older RHEL/CentOS systems.
rpm	RPM Package Manager: install, query, verify, and erase RPM packages.
pacman	Arch Linux package manager: install, update, remove, and search packages.
yay	AUR (Arch User Repository) helper that wraps pacman for AUR package management.
paru	Another AUR helper with additional features and a Rust implementation.
zypper	openSUSE package manager: install, update, remove packages and manage repositories.
emerge	Gentoo package manager (Portage): compile and install software from source.
nix	Nix package manager: purely functional package management for NixOS and other distros.
brew	Homebrew: package manager primarily for macOS, also available on Linux.
pip	Python package installer: install packages from PyPI.
pip3	Python 3 version of pip.
npm	Node.js package manager: install JavaScript packages from npmjs.com.
cargo	Rust package manager and build tool: compile and manage Rust crates.
gem	RubyGems package manager: install Ruby libraries and applications.
cpan	CPAN shell: download and install Perl modules from CPAN.

■ SHELL & SCRIPTING

40 commands

bash	Bourne Again SHell: the most common Linux shell, executes commands and scripts.
sh	POSIX-compatible shell (often symlinked to dash or bash on Linux).
zsh	Z Shell: feature-rich interactive shell with powerful completion and theming.
fish	Friendly Interactive SHell: user-friendly shell with autosuggestions and syntax highlighting.
echo	Print text or variable values to standard output.
printf	Format and print data, similar to C's printf; more control than echo.
read	Read a line of input from standard input into a variable.
export	Mark a shell variable for export to child processes as an environment variable.
unset	Remove a shell variable or function from the environment.
set	Display or set shell options and positional parameters.
env	Display or modify the environment for a command; print all environment variables.
printenv	Print the values of specified environment variables, or all if none specified.
source	Execute commands from a file in the current shell (same as the <code>.</code> built-in).
alias	Create a shortcut name for a longer command or set of commands.
unalias	Remove a previously defined alias.
history	Display the command history list for the current shell session.

fc	Fix command: edit and re-execute commands from history.
type	Indicate how a command name would be interpreted (builtin, alias, file, etc.).
which	Locate a command by searching the PATH and return its full path.
whereis	Locate the binary, source, and manual page files for a command.
command	Execute a command, bypassing shell functions and aliases.
builtin	Execute a shell built-in command, bypassing any defined function with the same name.
test	Evaluate conditional expressions; used in shell scripts for comparisons.
expr	Evaluate expressions (arithmetic, string, and logical) in a shell.
let	Evaluate arithmetic expressions in bash.
declare	Declare variables and give them attributes in bash.
local	Declare a variable as local in scope to a shell function.
return	Exit a shell function and return a value to the calling context.
exit	Exit the current shell or script with an optional exit status code.
trap	Catch and handle signals or other events in a shell script.
true	Do nothing and exit with a status of 0 (success). Useful in shell logic.
false	Do nothing and exit with a status of 1 (failure). Useful in shell logic.
xargs	Build and execute command lines from standard input.
tee	Read from stdin and write to both stdout and one or more files simultaneously.
exec	Replace the current shell process with a specified command, or redirect file descriptors.
eval	Concatenate arguments into a single command and execute it in the current shell.
getopts	Parse positional parameters (options) passed to a shell script.
shift	Shift positional parameters to the left by N positions in a script.
basename	Return just the filename component of a path (built-in or command).
mapfile	Read lines from stdin into an indexed array variable in bash.

■ TEXT PROCESSING

18 commands

grep	Search for lines matching a regex pattern in files or standard input.
sed	Stream editor for filtering and transforming text.
awk	Pattern-scanning and processing language for structured text.
cut	Remove sections (columns) from each line of input.
paste	Merge corresponding lines of files horizontally.
sort	Sort lines of text files lexicographically or numerically.
uniq	Report or omit repeated adjacent lines in sorted input.
tr	Translate or delete characters from standard input.
wc	Count lines, words, and bytes in files.
diff	Compare files line by line and show differences.
comm	Compare two sorted files line by line, showing lines unique to each.

cmp	Compare two files byte by byte and report the first difference.
strings	Find and print printable strings in binary files.
iconv	Convert text from one character encoding to another (e.g., UTF-8 to ISO-8859-1).
recode	Convert files between character sets and usages.
dos2unix	Convert Windows-style line endings (CRLF) to Unix-style (LF).
unix2dos	Convert Unix-style line endings (LF) to Windows-style (CRLF).
tput	Initialize terminal or query the terminfo database for terminal capabilities.

■ SYSTEM MONITORING & PERFORMANCE

26 commands

top	Real-time dynamic process viewer showing CPU, RAM, and process info.
htop	Enhanced interactive process viewer with color and mouse support.
atop	Advanced system and process monitor with historical recording capability.
glances	Cross-platform system monitoring tool with a web interface option.
iostat	Report CPU and I/O statistics for devices and partitions.
vmstat	Report virtual memory, processes, paging, block I/O, traps, and CPU activity.
mpstat	Report per-processor and global CPU statistics.
sar	Collect, report, or save system activity information (part of sysstat).
free	Display the amount of free and used memory (RAM and swap) in the system.
uptime	Show current time, how long the system has been running, and load averages.
pidstat	Report per-process CPU, memory, and I/O statistics.
pstat	Report system activity (varies by distribution).
dstat	Versatile resource statistics tool combining vmstat, iostat, netstat, etc.
perf	Performance analysis tool for Linux: profiling, tracing, and benchmarking.
valgrind	Memory debugging, memory leak detection, and profiling tool.
gprof	GNU profiler: analyze program performance from instrumented binaries.
time	Measure the total real, user, and system time consumed by a command.
systemd-analyze	Analyze systemd boot performance and critical chain.
slabtop	Display kernel slab cache information in real time.
numactl	Control NUMA (Non-Uniform Memory Access) policy for processes.
taskset	Set or retrieve the CPU affinity of a process (which CPUs it can run on).
cpufreq-info	Display CPU frequency scaling information.
sensors	Read hardware sensor data: temperatures, fan speeds, voltages.
acpi	Display ACPI information: battery, thermal zones, AC adapter status.
powertop	Analyze power consumption and optimize power usage on Linux systems.
turbostat	Report processor frequency, idle statistics, and power usage for x86 CPUs.

■ BOOT & INIT

18 commands

grub-install	Install the GRUB bootloader to a device.
update-grub	Regenerate the GRUB configuration file (Debian/Ubuntu).
grub-mkconfig	Generate a GRUB configuration file.
grub2-install	Install GRUB 2 bootloader (used on RPM-based distros).
systemctl	Manage systemd services and units: start, stop, enable, disable, status.
journalctl	Query and display messages from the systemd journal log.
init	Initialize the system (process 1); switch runlevels on SysV init systems.
telinit	Change the SysV init runlevel.
runlevel	Print the previous and current SysV runlevel.
reboot	Reboot the system.
poweroff	Power off the machine.
halt	Halt the system without powering off.
shutdown	Schedule a system shutdown or reboot with optional message to users.
wall	Send a message to all logged-in users' terminals.
dmesg	Print kernel boot messages and hardware detection output from the ring buffer.
dracut	Tool to generate an initramfs image for early boot.
mkinitcpio	Arch Linux tool to create an initial ramdisk environment (initramfs).
kexec	Load and boot a new kernel without going through the BIOS/UEFI firmware.

■ SECURITY & CRYPTOGRAPHY

29 commands

ssh-keygen	Generate, manage, and convert SSH authentication keys.
ssh-copy-id	Copy a public SSH key to a remote host's authorized_keys file.
ssh-agent	Hold private keys in memory for passwordless SSH authentication.
ssh-add	Add private keys to the ssh-agent for passwordless authentication.
openssl	Cryptography toolkit: TLS/SSL, key generation, certificates, encryption, hashing.
gpg	GNU Privacy Guard: public-key encryption, signing, and key management.
gpg2	Version 2 of GPG with improved architecture and smartcard support.
md5sum	Compute and check MD5 message digests for file integrity verification.
sha1sum	Compute and check SHA-1 checksums (deprecated for security use).
sha256sum	Compute and check SHA-256 checksums for file integrity.
sha512sum	Compute and check SHA-512 checksums (stronger than SHA-256).
base64	Encode or decode data in Base64 format.
uuencode	Encode a binary file into printable ASCII for transmission.
uudecode	Decode a uuencoded file back to binary.
chroot	Run a command or shell with a different root directory (isolation technique).

setenforce	Set SELinux to enforcing or permissive mode.
getenforce	Print the current SELinux enforcement mode.
sestatus	Show the status of SELinux: mode, policy, booleans.
aa-status	Show AppArmor module status and loaded profiles.
lynis	Security auditing and hardening tool for Linux/Unix systems.
auditctl	Control the Linux audit system: add/remove audit rules.
ausearch	Search the Linux audit logs for specific events.
fail2ban-client	Control the fail2ban intrusion prevention service.
nft	Administration tool for the nftables packet filtering framework.
cryptmount	Mount encrypted filesystems without needing root privileges.
veracrypt	Disk encryption software for creating encrypted volumes/containers.
age	Simple, modern file encryption tool with small key sizes.
pass	Standard Unix password manager using GPG encryption and git.
keepassxc	Cross-platform password manager storing passwords in an encrypted database.

■ DEVELOPMENT & DEBUGGING

44 commands

gcc	GNU C Compiler: compile C (and C++) source code into executables or object files.
g++	GNU C++ Compiler: compile C++ source code.
make	Build automation tool that reads a Makefile to compile projects.
cmake	Cross-platform build system generator that produces Makefiles or IDE projects.
gdb	GNU Debugger: debug programs, set breakpoints, inspect variables at runtime.
lldb	LLVM debugger: modern debugger with a Python API and Xcode integration.
clang	C/C++/Objective-C compiler from the LLVM project, known for helpful error messages.
python3	Run Python 3 scripts or launch the interactive Python interpreter.
python	Run Python scripts (may point to Python 2 or 3 depending on the system).
node	Execute JavaScript code using the Node.js runtime environment.
java	Run compiled Java programs using the JVM.
javac	Compile Java source files into bytecode.
rustc	Rust compiler: compile Rust source code into binaries.
go	Run, compile, test, and manage Go programs and modules.
perl	Run Perl scripts; powerful for text processing and system administration.
ruby	Run Ruby scripts or launch the interactive Ruby interpreter.
php	Run PHP scripts from the command line.
lua	Run Lua scripts; lightweight scripting language often embedded in applications.
git	Distributed version control system: track changes, branch, merge, collaborate.
svn	Apache Subversion: centralized version control system.

hg	Mercurial: distributed version control system (alternative to git).
strace	Trace system calls and signals; useful for debugging process behavior.
ltrace	Trace library function calls made by a process.
nm	List symbols from object files (functions, variables defined or used).
objdump	Display information about object files: disassembly, headers, symbols.
readelf	Display detailed information about ELF format binary files.
strip	Remove symbol table and debug information from binary files (reduces size).
ar	Create, modify, and extract from static library archives (.a files).
ld	GNU linker: link object files and libraries to create executables.
ldd	Print shared library dependencies of an executable or shared library.
pkg-config	Return information about installed libraries (compiler flags, linker flags).
ctags	Generate an index (tag) file of language objects for use by text editors.
cscope	Browse C/C++/Java source code interactively.
indent	Format C source code according to a specified style.
splint	Statically check C programs for security vulnerabilities and coding errors.
valgrind	Memory error detector and profiler: find leaks, invalid accesses, race conditions.
gprof	Profile execution of a program compiled with -pg flag.
gcov	Code coverage tool: measure which lines of code are executed during tests.
docker	Build, run, and manage containerized applications using Docker.
podman	Daemonless container engine; drop-in replacement for Docker.
kubect1	Command-line tool for controlling Kubernetes clusters.
ansible	IT automation tool: configure systems and deploy applications using YAML playbooks.
terraform	Infrastructure-as-code tool for provisioning cloud and on-premises resources.
vagrant	Build and manage reproducible development environments using virtual machines.

■ HELP & DOCUMENTATION

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man	Display the manual page for a command; the most important help system on Linux.
info	Read Info documents (GNU hypertext documentation system).
help	Display help for shell built-in commands.
whatis	Display a one-line description of a command from the manual page database.
apropos	Search the manual page database for commands matching a keyword.
tldr	Community-maintained simplified man pages with practical examples.
man -k	Search man page descriptions for a keyword (same as apropos).
--help	Most commands accept --help or -h flag to print a brief usage summary.
pinfo	Colorized info page reader with vi-like navigation.
manpath	Display the search path for manual pages.
makewhatis	Create or update the whatis database for man page searching.

update-alternatives	Maintain symbolic links for multiple versions of commands (Debian/Ubuntu).
xdg-open	Open a file or URL with the default application as configured by the desktop.

■ MISCELLANEOUS UTILITIES

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clear	Clear the terminal screen (move content off-screen).
reset	Reset the terminal to a sane state if display is corrupted.
tput reset	Reset terminal settings using the terminfo database.
script	Record a terminal session (all input and output) to a file.
screen	Terminal multiplexer: run multiple shell sessions in one terminal, detach/reattach.
tmux	Terminal multiplexer with split panes, windows, and session persistence.
byobu	Enhanced terminal multiplexer wrapper around screen or tmux.
xterm	Standard terminal emulator for X Window System.
xclip	Read from or write to X clipboard from the command line.
xsel	Read and write to the X selection (clipboard) from the command line.
xdotool	Simulate keyboard input and mouse activity; manipulate windows programmatically.
notify-send	Send desktop notifications from the command line.
zenity	Display GTK dialog boxes from shell scripts.
dialog	Display dialog boxes from shell scripts in terminal environments.
whiptail	Lightweight dialog box utility (uses ncurses).
bc	Arbitrary-precision calculator language with interactive and script modes.
dc	Reverse-polish desk calculator with arbitrary precision.
factor	Print the prime factorization of a number.
seq	Generate sequences of numbers.
yes	Repeatedly output a string (default 'y'); useful for automating confirmations.
cal	Display a calendar.
cowsay	Generate ASCII art of a cow (or other characters) saying a given message.
fortune	Print a random, hopefully interesting quote or saying.
figlet	Print large ASCII-art text banners from regular text.
lolcat	Concatenate files and apply rainbow coloring to the output.
banner	Print a large ASCII banner of given text (sysv-banner).
toilet	Display large colorful characters using various fonts (alternative to figlet).
sl	Display an ASCII art steam locomotive; runs when you mistype 'ls'.
cmatrix	Display a Matrix-like screensaver in the terminal.
asciinema	Record and share terminal sessions as lightweight text-based videos.
ffmpeg	Powerful multimedia framework for recording, converting, and streaming audio/video.
convert	ImageMagick tool to convert between image formats, resize, annotate images.

mogrify	ImageMagick in-place image transformation tool.
identify	Identify the format and characteristics of image files (ImageMagick).
display	Display an image or image sequence using ImageMagick.
feh	Lightweight image viewer and wallpaper setter.
sxiv	Simple X Image Viewer; minimal and fast.
mpv	Media player based on MPlayer/mplayer2; plays nearly any format.
vlc	VLC media player (command-line mode: cvlc or --no-video).
mplayer	Media player for Linux; predecessor to mpv.
sox	Sound exchange: process and convert audio files from the command line.
aplay	Play audio files using ALSA (Advanced Linux Sound Architecture).
arecord	Record audio using ALSA.
paplay	Play audio files through PulseAudio.
pactl	Control the PulseAudio sound server: list, set volume, switch sinks/sources.
wpctl	Control the WirePlumber/PipeWire session manager (volume, default devices).
pw-play	Play audio through PipeWire.
pw-record	Record audio through PipeWire.
amixer	Command-line ALSA mixer: adjust audio levels, mute, unmute.
alsamixer	ncurses-based ALSA mixer for interactive audio control.
mpd	Music Player Daemon: plays music in the background, controlled by clients.
mpc	Command-line client for MPD: add songs, control playback, manage playlists.
cava	Console-based Audio Visualizer for ALSA, PulseAudio, and PipeWire.
lp	Print a file to the default or specified printer.
lpq	Display the printer queue and status.
lprm	Remove print jobs from a printer queue.
lpadmin	Configure CUPS printer queues and classes.
cups	Common Unix Printing System: manage printers and print jobs.
xrandr	Set the size, orientation, and reflection of X screen outputs.
arandr	Graphical front-end for xrandr to configure multi-monitor setups.
xset	User preference utility for X: set screen blanking, keyboard repeat, etc.
setxkbmap	Set the keyboard layout and variant for X Window System.
xmodmap	Modify keymaps and pointer button mappings in X.
xbacklight	Set or get screen brightness via the backlight subsystem.
brightnessctl	Read and control device brightness (backlight, keyboard LEDs).
bluetoothctl	Interactive Bluetooth controller: pair, connect, and manage devices.
rfkill	Enable or disable wireless devices (Wi-Fi, Bluetooth) via the rfkill subsystem.
inotifywait	Wait for changes to files using the inotify API; useful in scripts.
inotifywatch	Gather filesystem access statistics using inotify.

lsmod	Show the status of currently loaded kernel modules.
modprobe	Add or remove kernel modules from the Linux kernel.
rmmod	Remove a module from the Linux kernel.
insmod	Insert a module into the Linux kernel.
modinfo	Show information about a Linux kernel module.
depmod	Generate a list of module dependencies (modules.dep).
sysctl	Configure kernel parameters at runtime via the /proc/sys interface.
dkms	Dynamic Kernel Module Support: build and install kernel modules automatically.
update-initramfs	Update or create the initial ramdisk for booting (Debian/Ubuntu).
dracut	Generate an initramfs image (used on RPM-based distributions).
locale	Get locale-specific information or display the current locale settings.
localectl	Query and change the system locale and keyboard layout settings (systemd).
tzselect	Interactively select a timezone.
hwclock	Read or set the hardware (BIOS/CMOS) clock.
ntpdate	Set the date and time via NTP (legacy; use timedatectl/chronyd).
chronyc	Command-line interface for the chrony NTP daemon.
ntpq	Query NTP servers and daemons.
logrotate	Rotate, compress, and mail log files automatically.
rsyslog	Rocket-fast syslog daemon for log management and forwarding.
logger	Make entries in the system log from the command line or shell scripts.
last	Show listing of last logged-in users.
lastb	Show bad (failed) login attempts.
faillog	Display or set failed login attempt counts.
utmpdump	Dump UTMP and WTMP files in raw format for analysis.
xdg-user-dirs-update	Update XDG user directories (Desktop, Downloads, etc.).
update-mime-database	Rebuild the MIME type database from installed XML files.
update-icon-caches	Rebuild icon theme caches.
fc-cache	Rebuild the font cache for fontconfig.
fc-list	List available fonts on the system.
fc-match	Match a font pattern and return the best matching font.
debconf	Debian configuration management system for packages.
dpkg-reconfigure	Reconfigure an already-installed Debian package.
tasksel	Debian/Ubuntu tool to install or remove groups of related packages.